

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: AI-Assisted Triage System of Middle-Acuity Patients in a Resource-Constrained Emergency Department: A 12-Week Local Validation and Shadow-Testing Pilot

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Statement of the Problem

(89 words)

The middle acuity band accounts for the majority of patients at Mercer's emergency department. Existing triage protocols cannot reliably distinguish patients in this band who will deteriorate from those who will not. Published AI triage models report strong predictive performance (3, 5) but are validated on single, well-resourced hospital datasets and rarely deployed prospectively (4). This project will develop and validate a predictive model on Mercer triage data, then evaluate it in shadow mode alongside nurse triage, to assess whether an AI-assisted system can improve identification of middle-band patients requiring up-triage.

Previous Work

Most recent work agrees that AI can sort ED patients better than standard triage, especially in the crowded middle band, but two problems recur and form the openings for a pilot at Mercer. Ten AI-triage papers are summarised below, followed by the two gaps they leave.

Summaries of the Literature

1. **Tyler et al. (2024), scoping review (1).** Tyler et al. conducted a scoping review of AI and ML applications in ED triage, searching EMBASE, Ovid MEDLINE and Web of Science for peer-reviewed primary studies from US journals, 2013 to 2023. Results demonstrated improvements in triage accuracy, disease identification and risk assessment, with reductions in under- and over-triage rates. The authors state that restricting the search to US publications limits generalisability, and that overconfident model outputs carry patient safety implications. This evidence base does not extend to non-Western or resource-constrained ED settings, limiting direct applicability to a Caribbean context.
2. **Yi et al. (2025), systematic review of prospective studies (2).** Yi et al. conducted a systematic review limited to prospective studies of AI-based ED triage, screening seven databases and synthesising the small number of studies that evaluated AI triage in the live clinical field rather than retrospectively. Reported discrimination varied widely across the included studies, with AUC values ranging from roughly 0.50 to 0.88, and few reported confidence intervals. The review shows that prospective evaluations are scarce, heterogeneous, and concentrated in well-resourced systems. It establishes, with systematic rigour, that prospective real-world evidence for AI triage remains thin, which is the central gap this project addresses.
3. **Levin et al. (2018), e-triage and the middle band (3).** Levin et al. developed an electronic triage (e-triage) system using machine learning on data from multiple EDs within a US health system and compared it against the ESI. Because more than 65% of visits fall in ESI level 3, the middle band, the study focused there: matching the ESI distribution, e-triage identified more than 10% of ESI level 3 patients (14,326) who warranted up-triage and carried markedly higher rates of critical care, emergency procedures and hospitalisation. The authors note the analysis is retrospective and state that further prospective validation is needed. This is the foundational evidence that the middle band is where mistriage concentrates, directly motivating the Mercer project.

4. **Taylor et al. (2025), real-world deployment (4).** Taylor et al. evaluated an AI-informed triage decision support tool across three EDs, comparing 83,404 pre-intervention and 91,244 post-intervention visits (174,648 total). After implementation, the share of critical-care patients correctly assigned to high-acuity triage (ESI 1 or 2) rose from 78.8% to 83.1%, and median time from arrival to initial care area fell from 12 to 8 minutes. Stated limitations include a quality-improvement design, a single health system and variable nurse compliance with tool recommendations. This is the only study reviewed that measures live prospective clinical impact, making it the strongest available evidence of real-world applicability.
5. **Phuttharaksa (2025), Bang Khla Hospital, Thailand (5).** Phuttharaksa evaluated ML classifiers for ED triage at Bang Khla Hospital, a resource-constrained community facility in Thailand, retaining 29,389 records after cleaning an initial 61,602, covering 2021 to 2024. A Random Forest achieved 98.67% overall accuracy, with an F1-score of 1.00 on the resuscitate and emergent levels. The author states that data is from a single hospital only, that EMR data quality and completeness may affect results, that key features including consciousness level and pain score were unavailable, and that the model has not been tested in real-world clinical settings. The near-perfect figures also suggest possible overfitting, so retrospective performance alone does not confirm clinical readiness, though the setting makes it the closest precedent to a Mercer pilot.
6. **De Freitas et al. (2020), Caribbean ED patient flow (6).** De Freitas et al. mapped patient flow in an emergency department in Trinidad and Tobago using non-participant observation, process mapping and informal interviews. They characterised the ED as a complex adaptive system and identified organisational factors that delay flow. The study is qualitative and process-focused, and tests no predictive or triage model. It establishes the Caribbean ED context directly relevant to Mercer, but leaves model-based re-stratification of acuity unaddressed.
7. **Liu et al. (2021), prospective ML triage (7).** Liu et al. developed a machine learning triage algorithm to detect patients needing critical care, then tested it in a prospective cohort that alternated standard and model-assisted triage over matched periods. This is one of few studies to evaluate an ML triage tool prospectively at the point of care rather than on retrospective data alone. The work was single-site and conducted in a well-resourced tertiary hospital. It did not target a middle-acuity band or a Caribbean population.
8. **Hinson et al. (2025), multisite deployment (8).** Hinson et al. reported outcomes from implementing an AI triage decision support tool across multiple emergency departments, focusing on triage equity. The tool was integrated into live nurse workflows and assessed for its effect on under- and over-triage across patient subgroups. The study is a real-world implementation rather than a controlled trial, and it runs within a single well-resourced US health system. It strengthens the evidence that AI triage can be deployed at scale, but says nothing about resource-constrained or Caribbean settings.
9. **El Ariss et al. (2024), resource allocation (9).** El Ariss et al. developed and validated a machine learning framework to predict the resources an ED patient will require, aiming to improve allocation during triage. They trained a suite of models on data from a US academic centre and reported standardised, bias-reduced resource forecasts. The work is constraint-focused in intent, but is validated on a single high-resource site and is wholly retrospective. It shows resource-aware triage is feasible, yet offers no evidence for a low-resource or prospective setting.
10. **Sitthiprawiat et al. (2025), Thai tertiary hospital (10).** Sitthiprawiat et al. developed and internally validated a model predicting ICU admission and resource use, using 163,452 ED visits from 2018 to 2022 at Maharaj Nakhon Chiang Mai Hospital in northern Thailand. An XGBoost model using vital signs, mode of arrival and multilingual free-text complaints outperformed the Canadian Triage and Acuity Scale (AUROC 0.917 versus 0.882). The study reports internal validation only, with no external testing and no prospective deployment. It is a strong non-Western, resource-aware precedent, but its single-site retrospective design leaves the same gaps as the others.

Identified Gaps

Gap 1: Single-site validation means there is no evidence of generalisability across different clinical populations or resource settings.

All the primary AI studies report results from a single site or health system. Tyler et al. (1) restrict their review to US publications and cite limited generalisability as a stated weakness. E-triage was developed within a single US health system (3). Phuttharaksa's model is built on one Thai hospital's records (5). Taylor's deployment is confined to one US health system (4). De Freitas et al. (6) document patient flow in a Caribbean ED but evaluate no predictive model. The more recent constraint-focused work repeats the pattern: El Ariss et al. (9) validate only internally on one US academic centre, and Sitthiprawiat et al. (10) report internal validation alone on a single Thai tertiary hospital. No study examines model performance on a Caribbean, resource-constrained, mixed-acuity ED. This gap is directly addressable by training and evaluating on Mercer's own data.

Gap 2: Prospective evaluation of AI triage tools in resource-limited settings is absent from the current literature.

The primary AI studies are largely retrospective (3, 5, 9, 10). Taylor et al. (4) and Liu et al. (7) provide the only prospective evidence, and Hinson et al. (8) the only real-world implementation, all within well-resourced systems in the United States and China. A systematic review of prospective AI triage studies confirms how few exist and how widely they vary, with none set in resource-limited environments (2). Workflow integration and clinician workload also remain unresolved barriers to adoption (11). These challenges cannot be evaluated from retrospective accuracy metrics alone. No study has conducted a prospective, point-of-care evaluation in a resource-constrained or Caribbean facility.

Systems Thinking and Workflow

Before specifying the solution, the current Mercer ED triage process was mapped from door to disposition to identify where an AI tool can realistically sit. Figure 1 shows the process with five candidate AI plug-in points and the shadow-mode validation loop.

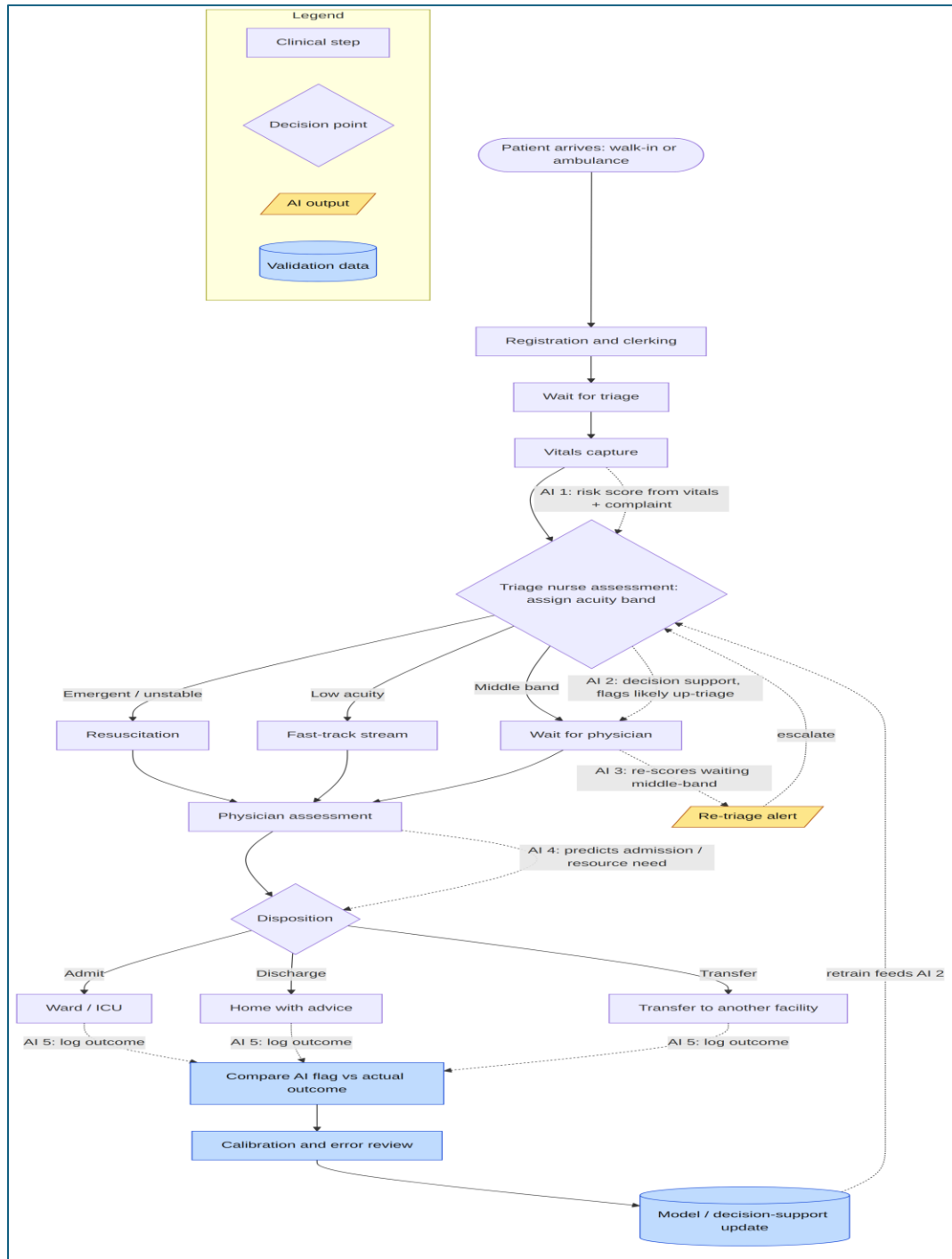


Figure 1. Current Mercer ED triage process. Dotted arrows mark AI plug-in points; yellow nodes are AI outputs and blue nodes the shadow-mode validation loop. The legend (top left) distinguishes clinical steps, decision points, AI outputs and validation data.

The five plug-in points are: (1) a risk score computed at vitals capture from vital signs and chief complaint; (2) decision support at the triage assessment that flags middle-band patients likely to need up-triage; (3) continuous re-scoring of middle-band patients waiting for a physician, raising a re-triage alert on deterioration; (4) admission and resource-need prediction at disposition; and (5) outcome logging that drives the shadow-mode validation loop, comparing each AI flag against the actual outcome and feeding calibration and error review back into the model.

Points 2 and 3 are the core of this project. The AI sits as decision support around the existing triage decision and the middle-band wait, not as a replacement for the triage nurse. Point 5 is how the pilot is evaluated without affecting care, and is the same loop described in Phase 3 of the proposed solution.

Constraints and Stakeholders

Workflow constraints

Three constraints in the current workflow must be respected by any AI design. Each is grounded in the literature and the Caribbean ED context.

1. **Vitals captured on paper before EHR entry introduce a delay.** Vital signs are often recorded on paper at the bedside and entered into the EHR later, and qualitative study of a Caribbean ED documents the organisational delays and information handoffs that slow patient flow (6). An AI tool cannot act on data that is not yet in the system. Mitigation: trigger inference only on EHR-confirmed records, or capture vitals at the point of care. Measurement: median paper-to-EHR lag during the pilot.
2. **Triage runs under high workload and frequent interruption.** A systematic review of the triage process finds that high workload, frequent interruptions and staffing pressures are leading threats to triage safety (11). Any output must be glanceable and add no data entry, or it will not survive a night shift. Mitigation: a passive flag inside the existing triage screen, with no new clicks. Measurement: added seconds per triage encounter, targeting zero.
3. **Vitals capture is inconsistent and often incomplete.** A high rate of missing vital signs at triage is well documented, varying with patient acuity and shift (12). The model must tolerate missing fields and free-text complaints rather than assume the clean inputs of a notebook. Mitigation: train and test with explicit missing-data handling and validate on realistically incomplete records. Measurement: model performance on records with at least one missing vital versus complete records.

Stakeholder map

Five clinical stakeholders and the concern each cares about most:

Stakeholder	Top concern
Triage nurse (Nurse-in-Charge)	Will it add work or slow me down, and can I override it?
ED consultant / physician	Reliability and accountability: missed deterioration and who is liable for an AI flag.
Porter / orderly	Clear, timely instructions on where and when to move patients.
Registration / admin clerk	Fast, accurate data capture without adding to queue time.
Patient advocate (for the patient)	Safety and equity: that no one is deprioritised by an opaque algorithm.

Proposed Solution

The proposed solution applies the deployment framework of Taylor et al. (4) to the Mercer context across four phases, designed around the constraints above.

Phase 1: Model development. A predictive model will be trained on Mercer's historical triage data using vital signs, chief complaint and patient disposition as inputs (4). Logistic regression and gradient boosting classifiers will be compared, with interpretability prioritised to support clinician trust and fit within the triage workflow (11), and with explicit handling of missing vitals per Constraint 3.

Phase 2: Local validation. The model will be evaluated on a held-out Mercer dataset. This directly addresses Gap 1. Poor performance at this stage will be identified before any clinical involvement.

Phase 3: Shadow-mode evaluation. The model will operate alongside live nurse triage for two to four weeks, logging predictions without influencing clinical decisions (4). Predictions will be compared against actual patient outcomes post-hoc. This directly addresses Gap 2 and provides the prospective, point-of-care evidence currently missing from the literature (2, 4).

Phase 4: Final deliverable. At week 12, the project will produce a locally validated model, shadow-mode performance metrics and a Board-level go or no-go recommendation.

Risk Analysis

This section draws on a 12-risk register covering AI-technical, operational, ethical and equity dimensions. The register, the top three risks in plain language, and a documented harm case follow. The design choices are informed by WHO guidance on the ethics and governance of AI for health (15) and by the ethical challenges of implementing machine learning in care (16).

Risk register

Likelihood and impact are rated Low, Medium or High for the pilot context. Each mitigation carries a measurable signal of success.

#	Risk	Category	Likelihood	Impact	Mitigation	Signal of success
1	Distribution shift. The model degrades as case-mix, seasonality (for example dengue surges) or documentation habits change	AI-technical	M	H	Local validation before use. Rolling performance and drift monitoring. Periodic recalibration	Monthly AUC and calibration remain within preset bounds on a rolling validation sample
2	Label leakage. Training on disposition or downstream actions teaches the model clinician behaviour rather than deterioration	AI-technical	M	H	Outcomes defined from objective events (ICU transfer, critical intervention) with a time gap after triage. Features audited for leakage	Leakage audit passes. Performance holds on a leakage-controlled holdout
3	Poor calibration. Overconfident scores mislead nurses	AI-technical	M	M	Calibration curves. Isotonic or Platt scaling. A confidence band is displayed, not a bare flag	Calibration slope near 1.0. Brier score below threshold
4	Missing or dirty inputs degrade predictions	AI-technical	H	M	Explicit missing-data handling. Output suppressed below a data-completeness threshold	Share of scores on incomplete records tracked. Complete against incomplete performance gap within tolerance
5	Alert fatigue. Too many false up-triage flags lead nurses to ignore the tool	Operational	H	H	Threshold tuned for high positive predictive value. Alert rate capped. Passive, non-interruptive display	Alert rate per shift and flag PPV monitored. Nurse dismiss rate trends down

#	Risk	Category	Likelihood	Impact	Mitigation	Signal of success
6	Workflow disruption. The tool adds clicks or slows triage	Operational	M	M	Passive flag inside the existing triage screen. No new data entry	Added seconds per triage near zero. Nurse usability rating
7	EHR lag. Paper-to-EHR delay means the model scores on stale data	Operational	H	M	Inference triggered only on EHR-confirmed data, or vitals captured at point of care	Median data-to-score latency. Share of scores on confirmed data
8	Silent failure and over-reliance. Staff who lean on the tool miss deterioration when it fails	Operational	M	H	A clear tool-unavailable state. Nurse-led triage remains primary. Downtime drills	Documented fallback adherence during simulated outages
9	Automation bias. Nurses defer to the model and under-triage when it is wrong	Ethical	M	H	Decision-support framing, with the nurse deciding. Brief rationale required on override. Shadow mode first	Nurse-model agreement against independent judgement tracked. Up-triage catches the model missed
10	Accountability and consent. Liability for an AI-influenced decision is unclear, and patients are unaware AI is involved	Ethical	M	M	Governance policy assigning clinician accountability. Patient-facing notice. Board sign-off	Accountability policy in place. Consent process audited
11	Training-data bias. Data skewed by age, sex, language or rural-urban access under-serves some groups	Equity	M	H	Subgroup performance audits. Reweighting. Cost and utilisation proxies avoided	Sensitivity equalised across age, sex and locale within tolerance
12	Proxy outcomes encode access inequity. Using admitted or resources-used as a label bakes in who historically received care	Equity	M	H	Labels based on clinical deterioration, not resource receipt. Equity review of the outcome definition	No systematic score gap by protected subgroup after acuity is controlled for

Top three risks

1. Alert fatigue (operational). A model that flags too many patients who are not deteriorating trains staff to ignore it. Once the flag is routinely dismissed, the tool misses the patient who matters. The Epic Sepsis Model failed in this way, as most of its alerts were false (13). The pilot will keep the alert rate low, favour precision over raw sensitivity, and monitor how often nurses dismiss the flag.

2. Distribution shift (AI-technical). A model can score well on Mercer's past data and still degrade as patients, seasons and recording habits change. The decay is gradual and easy to miss. The model will be validated locally before use and monitored month to month, with recalibration when performance slips.

3. Training-data bias (equity). If the data under-represents some groups by age, sex, language or location, the tool can serve them worse while average performance looks acceptable. A widely used US care-management algorithm did exactly this, under-serving Black patients because it used cost as a proxy for need (14). Performance will be measured per subgroup, and outcome labels based on resource receipt will be avoided.

Documented harm case: the Epic Sepsis Model

The clearest documented example is the Epic Sepsis Model, a proprietary sepsis early-warning tool deployed across hundreds of US hospitals. On independent external validation of 38,455 hospitalisations, it scored an AUC of 0.63 against a vendor-reported 0.76 to 0.83, missed about two-thirds of sepsis cases, and ran at 12% positive predictive value, which produced alert fatigue (13). The root causes, deployment without external validation, label leakage, low precision and proprietary opacity, map onto risks 1, 2 and 5 above, and are the reason this project validates locally and runs in shadow mode before influencing any decision. A full root-cause analysis is held in the project repository.

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Key Terms

Plain meanings for the terms used in this proposal, for any reader who is not from a machine learning background.

Term	What it means
Triage	Sorting patients by how urgently they need care, so the sickest are seen first.
Acuity	How severe or urgent a patient's condition is. High acuity means very sick or unstable.
Acuity band / level	The category a patient is placed in based on acuity. The middle band is moderate, neither clearly critical nor clearly minor.
ESI (Emergency Severity Index)	A common five-level triage scale used to rank how urgent a patient is.
KTAS (Korean Triage and Acuity Scale)	A similar five-level triage scale used in Korea.
Up-triage	Moving a patient to a higher urgency level than first assigned, because they are sicker than they looked.
Under-triage	Wrongly assigning a sick patient too low an urgency. This is dangerous.
Over-triage	Wrongly assigning a low-risk patient too high an urgency. This wastes resources.
Machine learning (ML)	A type of AI that learns patterns from examples in data rather than from fixed rules.
Model	The trained program that takes patient data and outputs a prediction.
NLP (Natural Language Processing)	AI that reads and understands free text, such as a nurse's written notes.
Clinical Decision Support (CDS)	A tool that gives clinicians a suggestion to help them decide. It does not replace them.
Accuracy	The share of predictions the model got right.
AUC (Area Under the Curve)	A score from 0.5 to 1.0 for how well a model separates two groups. Higher is better. 0.5 is a coin flip.
F1 score	A score from 0 to 1 that balances how often the model is right against how often it misses cases. Higher is better.
Retrospective	A study that uses data already collected in the past.
Prospective	A study that collects data going forward, in real time.
Shadow mode	Running the tool live but only logging its suggestions, without changing patient care, to see how it would have done.
Chief complaint	The main reason the patient says they came in.
Vital signs	Basic body measurements such as heart rate, blood pressure, temperature and oxygen level.
Disposition	What happens to the patient after the ED, for example admitted, discharged or sent to intensive care.